

Quiz: Portfolio Theory

Part I: Multiple Choice

1. In scenario analysis, the expected return on asset i is computed as:
 - A) The simple average of the returns across all scenarios
 - B) The return in the most likely scenario
 - C) The probability-weighted average of the returns across all scenarios
 - D) The highest return the asset can achieve
2. The holding period return (HPR) on a share is measured as:
 - A) The price change plus any income received, divided by the beginning price
 - B) The price change divided by the ending price
 - C) The income received divided by the beginning price
 - D) The price change divided by the beginning price, ignoring any income
3. For a series of returns that varies from period to period, the geometric mean return is:
 - A) Always larger than the arithmetic mean return
 - B) Exactly equal to the arithmetic mean return
 - C) Larger or smaller depending on the sign of the returns
 - D) Always smaller than the arithmetic mean return
4. Diversification reduces portfolio risk most effectively when the assets held:
 - A) Have identical expected returns
 - B) Are less than perfectly positively correlated with each other
 - C) Are perfectly positively correlated with each other
 - D) All belong to the same industry
5. Which of the following risks CANNOT be eliminated through diversification?
 - A) The risk that a firm loses a major lawsuit
 - B) The risk that a firm's CEO dies unexpectedly
 - C) The risk that one industry suffers a downturn
 - D) The risk arising from the business cycle

6. A zero-variance portfolio can be constructed from two risky assets when the correlation between them is equal to:
- A) -1
 - B) 0
 - C) $+0.5$
 - D) $+1$
7. In the utility function $U = E(r) - \frac{1}{2}A\sigma^2$, an investor with $A = 0$ is:
- A) Risk-averse, and requires a risk premium to hold a risky asset
 - B) Risk-loving, and gains utility from taking on risk
 - C) Risk-neutral, and ranks portfolios by expected return alone
 - D) Unable to rank portfolios at all
8. An investor places a fraction y of their capital in a risky portfolio P and the remainder in the risk-free asset. The standard deviation of the resulting complete portfolio is:
- A) σ_P , regardless of the value of y
 - B) $y \cdot \sigma_P$
 - C) $y^2 \cdot \sigma_P$
 - D) $y \cdot \sigma_P + (1 - y) r_f$
9. The slope of the Capital Allocation Line measures:
- A) The additional expected return earned per unit of additional risk, i.e. the Sharpe ratio
 - B) The investor's degree of risk aversion
 - C) The variance of the risky portfolio
 - D) The proportion of wealth held in the risk-free asset
10. Which statement about the minimum-variance frontier is correct?
- A) Every portfolio lying on the frontier is efficient
 - B) The global minimum variance portfolio offers the highest expected return of all risky portfolios
 - C) Risk-averse investors prefer the portfolios lying below the global minimum variance portfolio
 - D) Only the portion of the frontier above the global minimum variance portfolio is the efficient frontier

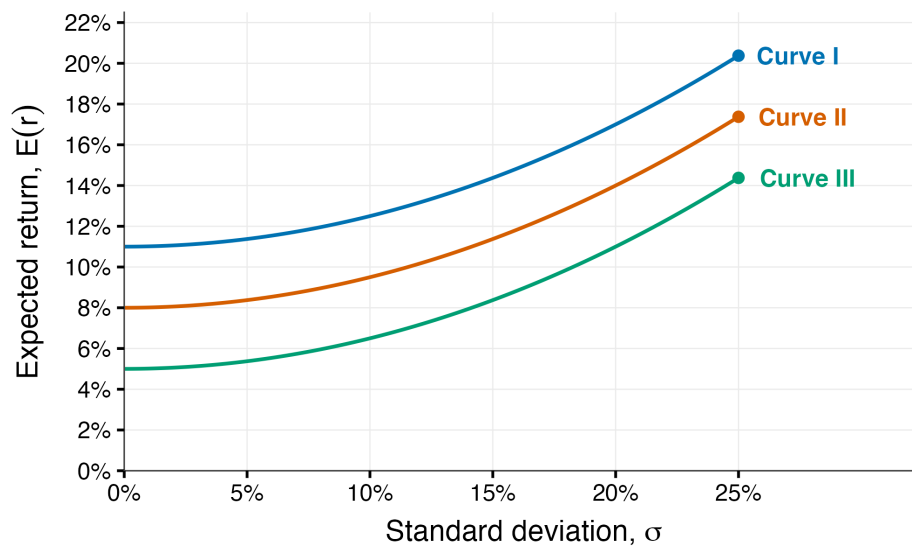


Figure 1: Indifference curves – same curvature.

11. Figure 1 shows three indifference curves. Which curve gives the investor the **highest utility**?
- A) Curve I
 - B) Curve II
 - C) Curve III
 - D) They all give the same utility, since the risk aversion coefficient is the same

Questions 12–14 refer to Figure 2 below.

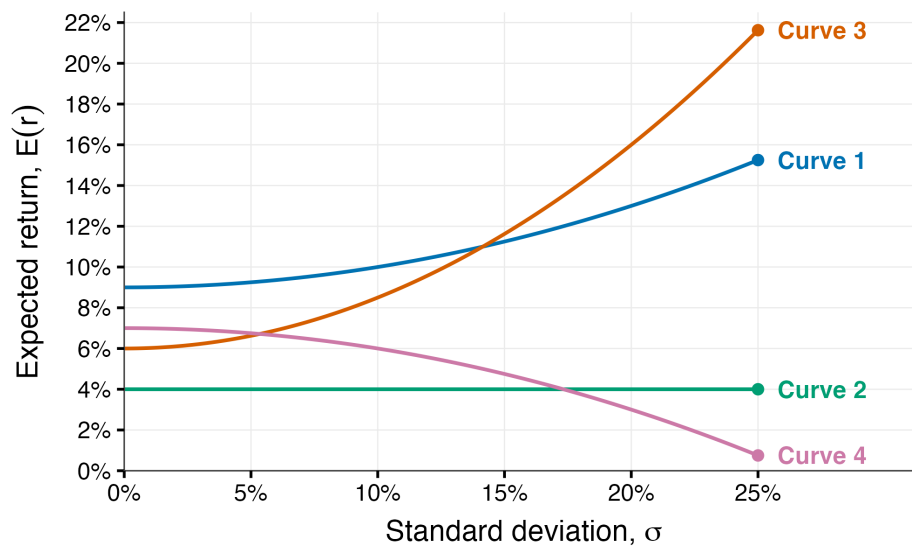


Figure 2: Indifference curves for four investors with different risk aversion and utility scores.

12. Which curve belongs to the **most risk-averse** investor, i.e. the investor with the largest risk aversion coefficient A ?
 - A) Curve 1
 - B) Curve 2
 - C) Curve 3
 - D) Curve 4
13. Which curve represents the **highest utility score** U ?
 - A) Curve 1
 - B) Curve 2
 - C) Curve 3
 - D) Curve 4
14. Which curve belongs to a **risk-neutral** investor?
 - A) Curve 1
 - B) Curve 2
 - C) Curve 3
 - D) Curve 4

Part II: Calculation Questions

15. Scenario analysis and portfolio choice.

You are evaluating two funds. Their returns depend on the state of the economy next year as follows:

State of the Economy	Probability	Fund A	Fund B
Recession	0.25	−12%	−4%
Normal growth	0.50	10%	8%
Boom	0.25	28%	16%

- Compute the expected return of each fund.
- Compute the variance and standard deviation of each fund.
- An investor has a risk aversion coefficient of $A = 4$. Using the utility score $U = E(r) - \frac{1}{2}A\sigma^2$, determine which fund the investor prefers.
- Which fund would a **risk-neutral** investor choose? Explain why the answer differs from part (c).

16. The complete portfolio and the Capital Allocation Line.

You manage a risky portfolio P with an expected return of 14% and a standard deviation of 20%. The risk-free rate is 5%.

- A client invests 60% of their wealth in your fund and the remaining 40% in treasury bills. Compute the expected return and standard deviation of the client's complete portfolio.
- Compute the slope of the Capital Allocation Line and state what it measures.
- A second client has a risk aversion coefficient of $A = 3$. Compute the optimal weight y^* in the risky portfolio, and the expected return and standard deviation of their optimal complete portfolio.
- A third client has $A = 6$. Without redoing the full calculation, state whether their y^* is higher or lower than in part (c), and explain the intuition.